



NeuralGCM: Hybrid Physics-AI for Weather and Climate

With Additional Tropical Cyclone Case Studies

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Abstract

Weather forecasting has long been essential to society. The quiet revolution of numerical weather prediction transformed forecasting through physics-based models, while recent advances in machine learning have opened a new phase in data-driven prediction. NeuralGCM brings these approaches together as a hybrid physics-AI model for weather and climate. The overarching aim of this project is to evaluate NeuralGCM as a weather and climate model, by considering both benchmark forecast accuracy and physically meaningful weather behaviour. This project reproduces key results from the original NeuralGCM paper and supports its broad conclusion that NeuralGCM provides competitive medium-range forecast accuracy. However, the reproduction is not exact, with several accuracy metrics weaker than in the original study, most notably for the 500 hPa geopotential height. Nevertheless, NeuralGCM performs well in the Hurricane Laura case study and reproduces annual tropical cyclone track distributions. To test the model further, three additional tropical cyclone case studies are analysed: Gulab and Shaheen, Faraji, and Seroja. These are chosen to assess re-intensification, track curvature, and the Fujiwhara effect respectively. The extension results are mixed. The cyclone track is partly captured, with the deterministic models performing best for Faraji. However, Shaheen's re-intensification is missed, and although Seroja's early track is captured, the tracks deteriorate during or shortly after the Fujiwhara interaction. Across all cases, NeuralGCM underestimates the intensity and produces weaker, shallower vortex structures. Overall, NeuralGCM shows promise as a hybrid physics-AI weather and climate model, but the results show that it inherits some limitations associated with data-driven forecasting, like underestimating intensity in tropical cyclones.

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1 Introduction

Weather forecasting is essential for society, guiding decisions from everyday choices, like taking a coat, to major preparations by governments and emergency services for extreme weather events [1, 2]. Modern weather prediction can be traced to Lewis Fry Richardson, who theorised in 1922 that the atmosphere could be forecast by solving the equations of physics [3]. This became possible in 1950, when Jule Charney, Ragnar Fjørtoft and John von Neumann produced the first successful numerical weather forecast on the Electronic Numerical Integrator and Computer (ENIAC) [4, 5]. The same approach gave rise to General Circulation Models (GCMs), which simulated the large-scale circulation of the atmosphere and later became central to climate prediction [6].

Numerical Weather Prediction (NWP) remains the foundation of operational forecasting and the main benchmark for medium-range weather prediction. Over the past 70 years, models have improved steadily through advances in computing power, numerical methods, data assimilation, and representations of atmospheric physics. This progress has often been described as the quiet revolution of NWP [7].

This progress relies on observational data from satellites, surface stations, and ocean observing systems. These observations are combined with model forecasts through data assimilation to improve the initial atmospheric state used in operational forecasting [8, 9]. The same approach is used to reconstruct past atmospheric states in reanalysis datasets, such as ERA5, the fifth generation atmospheric reanalysis produced by the European Centre for Medium Range Weather Forecasts [10]. These datasets made the rapid development in data-driven weather prediction possible. Data-driven models use Machine Learning (ML) to predict future atmospheric states from past states, rather than solving the physics equations directly. Since most data-driven models are trained on reanalysis data, their performance depends on this observational, data assimilation, and data-sharing infrastructure, much of which is collected and maintained by public institutions [11].

Early work by Ryan Keisler in 2022 showed that graph neural networks could produce competitive global forecasts [12]. This was followed by several high-performing models, including Pangu Weather, AIFS, GenCast, and GraphCast [13, 14, 15, 16]. These models have demonstrated impressive accuracy at medium-range lead times, often producing forecasts in seconds or minutes once trained, making them much faster at inference than current NWP models.

However, purely data-driven models have important limitations. Because they learn from historical data, their ability to generalise to rare or climate-shifted events remains uncertain [17, 18, 19]. This is particularly important for extreme events, which are sparse in the training data but have the greatest impact on people across the globe. Recent studies show that while data-driven models can be competitive for some extreme event forecasts, their performance is less reliable for the rarest and most intense cases, where they often underestimate the frequency and intensity of record-breaking events relative to physics-based systems [20, 21]. A further challenge is long-term stability, as many data-driven weather models can drift or become unstable when rolled out far beyond weather timescales, limiting their direct use for climate simulation [22, 23].

This motivates a hybrid approach that combines NWP and data-driven methods. NeuralGCM [24] is a hybrid model that combines a physical dynamical core for large-scale atmospheric processes with a ML component for sub-grid parametrisations. As one of the first hybrid models of its kind, NeuralGCM is a test case for whether physical modelling and ML can be reliably combined. Therefore, it requires rigorous evaluation to assess whether its reported performance is robust, physically reliable, and suitable for further development.

1.1 Research Scope

The overarching aim of this study is to evaluate whether NeuralGCM is a good weather and climate model. Following Allan Murphy’s definition of forecast goodness, goodness is not understood through accuracy alone, but through the relationship between forecasts, observations, and their value to users [25]. Therefore, in this study NeuralGCM is assessed by both its benchmark forecast accuracy and its ability to represent physically meaningful weather behaviour.

The primary focus of this research is to reproduce key results from the original NeuralGCM paper [24]. This is done in three stages. First, several accuracy metrics are calculated and performance is compared with other data-driven and NWP forecasts. Second, Hurricane Laura is used as a case study to test medium-range weather forecasting. Third, the climate behaviour of the model is examined through a one-year forecast from which annual tropical cyclone tracks are calculated. Together, these experiments assess whether the reported accuracy of NeuralGCM can be reproduced.

Following this reproduction, the research examines three 2021 tropical cyclone cases in more detail: Gulab and Shaheen, Seroja, and Faraji. While existing work has considered broad tropical cyclone behaviour in NeuralGCM, detailed intensity measures and vortex structure remain less well explored [26]. The Seroja case also allows this study to examine binary vortex interaction through the Fujiwhara effect, which, to the best of the author’s knowledge, has not previously been assessed using NeuralGCM or other data-driven weather models. The extension analysis aims to answer the following questions:

- Does NeuralGCM capture the cyclone tracks relative to the IBTrACS best track?
- Does NeuralGCM reproduce the observed cyclone intensity measures and vertical coherence?
- Does NeuralGCM capture the key dynamical challenge in each case: Gulab and Shaheen re-intensification, Faraji track curvature, and the Seroja-Odette Fujiwhara interaction?

1.2 Report Layout

The remainder of this report is structured as follows. Section 2 introduces NeuralGCM, evaluation metrics, datasets, and the tropical cyclone explanation relevant to this study. Section 3 describes the methodology used for the reproduction and extension experiments. Sections 4 and 5 present and discuss the results, then Section 6 concludes.

2 Background

2.1 NeuralGCM Model

NeuralGCM is a hybrid model that combines a differentiable physical dynamical core with learned neural network parametrisations, as shown in Figure 1a. The model is initialised from an atmospheric state y_t , a set of ERA5 fields at time t , including wind, temperature, humidity, geopotential height and surface pressure. It also receives prescribed forcings F_t , including sea surface temperature and sea ice concentration. These forcings provide boundary and radiative conditions but are not freely predicted by the model. In stochastic versions, random noise z_t is also supplied to represent forecast uncertainty and generate different ensemble realisations.

All these inputs are passed to a learned encoder, which maps the ERA5 pressure-level fields into the model’s internal state x_t on σ coordinates. Each forecast step then combines two components. The first is the dynamical core, which is the physics component of the model. It advances the resolved atmospheric flow by solving the hydrostatic primitive equations. These equations describe the evolution of wind, temperature, pressure and moisture under advection, pressure gradients, gravity and the Coriolis force.

The second component is the learned physics module (see Figure 1b), which is the ML component of NeuralGCM. It parametrises physical processes that are not resolved explicitly by the dynamical core, including radiative transfer, clouds and precipitation. Rather than solving these processes from first principles, it learns their net effect on the model state from data. Architecturally, the module is a column-wise fully connected neural network with residual connections. Shared weights are then applied independently to each vertical column. The output is a set of predicted rates of change, or tendencies, of the prognostic variables over one model time step.

The tendencies from the dynamical core and learned physics module are combined and advanced using a semi-implicit time integration scheme. This produces the next model state x_{t+1} , which is then fed back into the model to generate subsequent forecast steps. After N steps, a learned decoder maps the model state from σ coordinates back to standard pressure-level forecast variables.

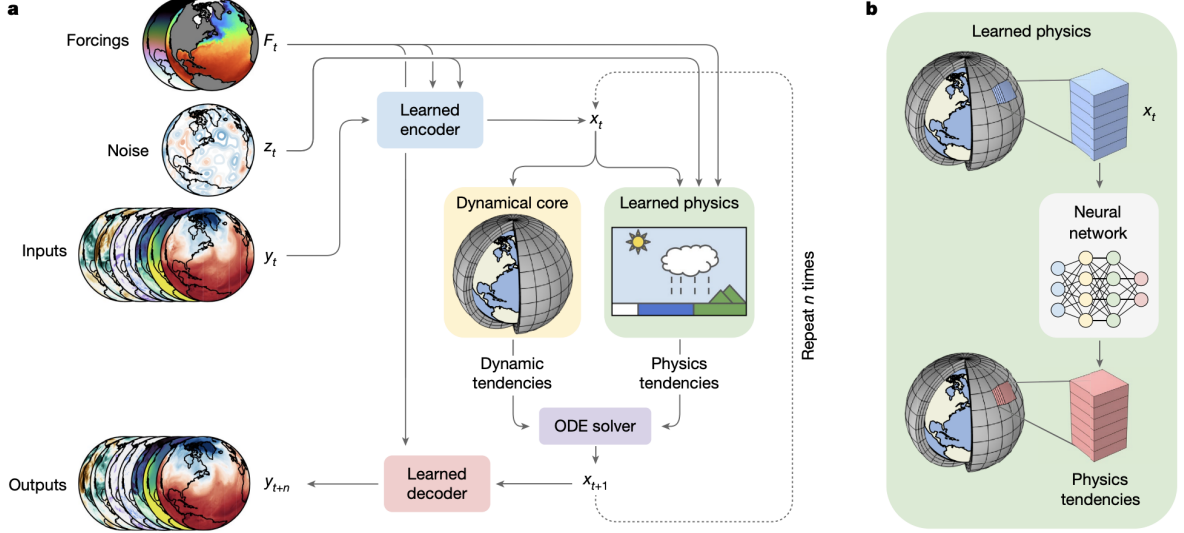


Figure 1: Overview of the NeuralGCM architecture. **(a)** Full model workflow from forcings, noise and atmospheric inputs through the encoder, dynamical core, learned physics module, ODE solver and decoder. **(b)** Column-wise learned physics module, where a neural network predicts physics tendencies from the model state. Figure taken from [24], licensed under CC BY 4.0

A key feature of NeuralGCM is that it is trained end-to-end on ERA5 reanalysis data. This is possible because the physical dynamical core is differentiable, so the encoder, learned physics module and decoder can be optimised through complete forecast rollouts rather than being trained separately. During training, the rollout length is gradually increased from six hours to five days, helping the model learn stable multi-step forecasts.

The deterministic model is trained in two stages, primary training and decoder fine-tuning. In the primary stage, the objective is a weighted sum of five mean squared error loss terms, including accuracy and bias terms evaluated in data space and model space. Decoder fine-tuning is then applied and only the decoder parameters are updated. This removes spectral artefacts caused by truncation errors when transforming from spherical harmonic to grid-point space. The stochastic model is trained using the Continuous Ranked Probability Score (CRPS).

2.2 Evaluation Metrics

This section defines the evaluation metrics used to assess NeuralGCM. Let $X(t \rightarrow t + \tau)$ denote a forecast initialised at time t and evaluated at lead time τ , and let $Y(t + \tau)$ denote the corresponding reference value.

1. Root Mean Square Error (RMSE):

RMSE gives the typical size of forecast errors, with larger errors penalised more heavily.

$$\mathcal{E}_{RMSE}(\tau, v, p) := \sqrt{\frac{1}{|T|} \sum_{t \in T} \|X(t \rightarrow t + \tau, v, p) - Y(t + \tau, v, p)\|_{nodal}^2}.$$

where v is the variable, p is the pressure-level, and T is the set of initial times. The nodal norm is area weighted

$$\begin{aligned} & \|X(t \rightarrow t + \tau, v, p) - Y(t + \tau, v, p)\|_{nodal}^2 \\ &:= \frac{1}{IJ} \sum_{i=1}^I \sum_{j=1}^J w(i) |X(t \rightarrow t + \tau, v, p, i, j) - Y(t + \tau, v, p, i, j)|^2. \end{aligned}$$

Here i and j are latitude and longitude indices, I and J are the corresponding grid sizes, and $w(i) \propto (\sin \phi_{i+0.5} - \sin \phi_{i-0.5})$ is the latitude area weight.

2. Root Mean Square Bias (RMSB):

RMSB gives the area weighted size of the time mean forecast error, measuring the magnitude of systematic spatial bias regardless of sign.

$$\mathcal{E}_{RMSB}(\tau, v, p) := \sqrt{\frac{1}{IJ} \sum_{i=1}^I \sum_{j=1}^J w(i) [\mathcal{E}_{bias}(\tau, v, p, i, j)]^2}.$$

3. Bias:

Bias estimates the persistent difference between forecasts and the reference. It can be positive or negative, depending on whether the forecast is too high or too low on average.

$$\mathcal{E}_{bias}(\tau, v, p, i, j) := \frac{1}{|T|} \sum_{t \in T} [X(t \rightarrow t + \tau, v, p, i, j) - Y(t + \tau, v, p, i, j)].$$

4. Continuous Ranked Probability Score (CRPS):

CRPS evaluates probabilistic forecasts by comparing the forecast distribution with the reference. Lower values indicate better probabilistic performance. For scalar random variables, CRPS can be written as

$$CRPS(U, V) = \mathbb{E}|U - V| - \frac{1}{2} \mathbb{E}|U - U'|.$$

where V is the reference random variable, and U and U' are independent forecasts from the predicted distribution. For M ensemble members $(X^{(1)}, \dots, X^{(M)})$, this is estimated as

$$\begin{aligned} \mathcal{E}_{CRPS}(\tau, v, p) &:= \frac{1}{|T|} \sum_{t \in T} \frac{1}{IJ} \sum_{i=1}^I \sum_{j=1}^J w(i) \left[\frac{1}{M} \sum_{m=1}^M |X^{(m)}(t \rightarrow t + \tau, v, p, i, j) - Y(t + \tau, v, p, i, j)| \right. \\ &\quad \left. - \frac{1}{2M(M-1)} \sum_{\substack{m,k=1 \\ m \neq k}}^M |X^{(m)}(t \rightarrow t + \tau, v, p, i, j) - X^{(k)}(t \rightarrow t + \tau, v, p, i, j)| \right]. \end{aligned}$$

5. Spread Skill Ratio (SSR):

Following the original NeuralGCM paper, SSR is the ratio of ensemble spread to the RMSE of the ensemble mean. Values close to one indicate that ensemble spread is consistent with ensemble mean forecast error. It is calculated as

$$\begin{aligned}
SSR &= \sqrt{\frac{S^2}{\epsilon^2 - S^2/N}}, \\
S^2 &= \frac{1}{|T|} \sum_{t \in T} \frac{1}{N-1} \sum_{n=1}^N \left(X_t^{(n)} - \mu_t \right)^2, \\
\epsilon^2 &= \frac{1}{|T|} \sum_{t \in T} (Y_t - \mu_t)^2, \\
\mu_t &= \frac{1}{N} \sum_{n=1}^N X_t^{(n)}.
\end{aligned}$$

2.3 Tropical Cyclones

Tropical cyclones are rotating, warm-core low-pressure systems that form over oceans in the tropics. They are known as ‘hurricanes’ in the Atlantic and eastern North Pacific and tropical cyclones in the Indian Ocean and South Pacific. Although basin conditions differ, they both share broadly similar underlying dynamics. Their rotation is set by the Coriolis effect, giving anticlockwise circulation in the Northern Hemisphere and clockwise circulation in the Southern Hemisphere [27].

Cyclone formation begins when a pre-existing disturbance develops over oceans in the tropics, where the sea surface supplies heat and moisture to the lower atmosphere. Warm, moist air converges near the surface and rises, forming deep convective clouds. Condensation releases latent heat, warming the core and lowering surface pressure. The pressure fall strengthens near-surface inflow, which feeds further convection. If the surrounding atmosphere is moist, vertical wind shear is weak, and the Coriolis force is strong enough to support rotation, this feedback can organise the disturbance into a vertically coherent warm-core vortex. As the cyclone intensifies, subsidence near the centre can suppress cloud formation and create a relatively calm region, known as the eye. The surrounding eyewall contains the strongest winds and deepest convection [28, 29].

Nearby tropical cyclones may undergo binary interaction through the Fujiwhara effect, in which two cyclonic vortices mutually advect and rotate about a common centre [30]. This can modify storm track, translation speed, and intensity, with close interactions potentially leading to vortex merger or absorption. Although separation thresholds are approximate, this study uses the following scales: 1450 km for broad binary interaction [31], 650 km for Fujiwhara-dominated relative rotation [32], and 300 km for a possible merger [33].

Once formed, the two main properties of interest in this project are track and intensity. The track is the path followed by the cyclone eye through time. In this project, this is estimated in two ways. The first method uses the local minimum in Mean Sea Level Pressure (MSLP). MSLP is the atmospheric pressure reduced to sea level, and its minimum marks the surface low associated with the cyclone. The second method uses the local maximum in 850 hPa cyclonic relative vorticity. This measures the rotation of the horizontal wind field, and its maximum marks the strongest part of the lower-tropospheric vortex. The 850 hPa level is used for vorticity because it captures the lower-tropospheric circulation while reducing the influence of near-surface friction.

Intensity describes the strength of the cyclone circulation and is usually defined operationally by maximum sustained 10 m wind [34]. Minimum MSLP is also reported as an intensity indicator, but intensity depends more broadly on pressure, wind field, size, and vertical structure

[29]. Therefore, this project assesses intensity using 850 hPa geopotential height differences, near-centre wind strength, and cyclonic relative vorticity structure, since stronger cyclones are expected to have lower MSLP, stronger winds, larger low-level cyclonic relative vorticity, and a more vertically coherent vortex [27].

Cyclones are useful stress tests for data-driven weather models because they are rare, intense, and high-impact events. Recent evaluations show that these models can produce competitive track forecasts but are weaker on intensity [35]. NeuralGCM is an interesting test case for whether hybrid design improves cyclone structure and intensity. Table 1 summarises the selected cyclones to test this.

Sub-Region	Name	Maximum SSHWS	Form Date	Dissipated Date
North Atlantic	Laura	Category 4	20/08/2020	29/08/2020
Bay of Bengal and Arabian Sea	Gulab and Shaheen	Category 1	24/09/2021	04/10/2021
South West Indian Ocean	Faraji	Category 5	04/02/2021	15/02/2021
Australian Region	Seroja	Category 1	05/04/2021	12/04/2021
Australian Region	Odette	N/A	03/04/2021	10/04/2021

Table 1: Cyclone case study events by sub-region. Maximum intensity is given using Saffir-Simpson Hurricane Wind Scale (SSHWS) categories where 5 indicates the highest intensity [36]. Odette is included because it interacts with Seroja during the Fujiwhara interaction, but it is not a case study on its own.

Hurricane Laura was chosen because it appears in the original NeuralGCM paper. The three extension cases were chosen because each presents a different dynamical challenge. Gulab and Shaheen test whether NeuralGCM can maintain a remnant circulation after landfall and represent later re-intensification over the Arabian Sea. Faraji tests whether the model can capture track curvature and vortex organisation along a recurving path. Seroja tests whether NeuralGCM can represent binary interaction with Tropical Storm Odette through the Fujiwhara effect.

The final two cases are deliberately selected from the Southern Hemisphere. Most model evaluations and cyclone case studies focus on Northern Hemisphere cyclones, where conventional observations are generally denser, while Southern Hemisphere ocean basins rely more heavily on satellite observations. These cases test whether data-driven forecast models generalise beyond more commonly evaluated, observation-rich regions [37].

2.4 Data

This project used data from several sources, with different access routes depending on the task. NeuralGCM forecast generation, post-processing, and limited evaluation were run on JASMIN, the United Kingdom’s data analysis facility for environmental science [38]. The main data sources were as follows:

- **ERA5 reanalysis.** ERA5 is the fifth generation ECMWF reanalysis product, combining observations with a numerical weather prediction model to estimate the historical atmosphere from 1940 onwards [10]. It was used for NeuralGCM inputs, surface forcings, reference fields, and cyclone diagnostics, and was accessed through:

- **Centre for Environmental Data Analysis (CEDA) [39] archive on JASMIN:** ERA5 model-level atmospheric fields and surface fields used for NeuralGCM forecast generation.
 - **Copernicus Climate Data Store (CDS) [40]:** Used to access cloud ice and cloud liquid water fields that were unavailable on CEDA on JASMIN.
 - **WeatherBench2 [41]:** Used to access standardised ERA5 reference fields for forecast evaluation, climatological comparison, and ERA5 cyclone tracking at all model resolutions.
- **Forecast products.** WeatherBench2 also provided operational NWP forecasts from IFS HRES and IFS ENS, and data-driven forecasts from GraphCast and Pangu Weather, used to compare NeuralGCM forecast performance.
 - **Cyclone reference tracks.** Cyclone track data were taken from the International Best Track Archive for Climate Stewardship (IBTrACS) version 4r01 [42], which provides global tropical cyclone best track position and intensity estimates from multiple agencies.

3 Methodology

This section describes the shared NeuralGCM forecast generation and cyclone tracking workflows, followed by case-specific details.

3.1 Forecast Generation Workflow

NeuralGCM forecasts followed a shared workflow, although the model resolution, initialisation dates, forecast lengths, output variables and forcing mode varied between experiments. The workflow has two stages:

1. Pre-processing

NeuralGCM input files were prepared from ERA5 atmospheric and surface fields. The three-dimensional atmospheric fields were temperature, specific humidity, zonal wind, meridional wind, and geopotential height. These were obtained from the CEDA archive on JASMIN as ERA5 model-level data, then vertically interpolated onto the required pressure levels, horizontally regridded to the NeuralGCM grid, renamed using NeuralGCM variable conventions and saved as daily files. The two-dimensional surface fields were sea surface temperature and sea ice cover. These were also obtained from CEDA and regridded to the NeuralGCM grid, with residual missing values filled using nearest-neighbour interpolation.

Specific cloud ice water content and specific cloud liquid water content were downloaded as pressure level fields, then converted to the same coordinate and variable conventions as the other NeuralGCM inputs and regridded to the target NeuralGCM grid. The original NeuralGCM paper does not specify how residual missing values in these cloud inputs were handled, so any remaining missing values were treated as cloud-free points and filled with zero before the cloud fields were merged with the atmospheric and surface input streams.

2. Running NeuralGCM

The NeuralGCM models used were the 0.7° deterministic model, the 1.4° deterministic model, and the 1.4° stochastic model. Stochastic forecasts were generated as independent seeded realisations and then merged along a member dimension. Forecasts used either persistent surface forcings, where Sea Surface Temperature (SST) and sea ice cover were taken from one day before forecast initialisation and held fixed throughout the forecast; or dynamic surface forcings, where ERA5 sea surface temperature and sea ice cover was updated every six hours through the forecast period.

All NeuralGCM forecasts were run on JASMIN using the ORCHID GPU cluster, which has 16 nodes with NVIDIA A100 GPUs. Full implementation details and scripts are provided in the accompanying repository.

3.2 Cyclone Tracking Workflow

A common cyclone tracking workflow was used throughout the project. This section describes the method and the two tracker types.

Cyclone tracking was performed with `TempestExtremes` in two steps [43]. First `DetectNodes` identified candidate cyclone centres independently at each time step. Second, `StitchNodes` connected them into continuous tracks. The two tracker types differed in the field used to define candidate centres and in the detection constraints. For both trackers, `StitchNodes` linked candidates using constraints on maximum step distance, minimum track duration, allowed temporal gaps, and minimum path length. Parameter values were set separately for each case, with full case-specific configuration files provided in the accompanying repository.

- **Mean Sea Level Pressure (MSLP) tracker**

The MSLP tracker identifies candidate cyclone centres as local minima in sea level pressure. In `DetectNodes`, these minima are filtered using a closed contour pressure test and a 10 m wind threshold within a specified radius, with an additional track-level wind threshold applied where required.

- **Vorticity tracker**

The vorticity tracker identifies candidate cyclone centres as local maxima in 850 hPa cyclonic relative vorticity. Relative vorticity is sign-adjusted where necessary so that cyclonic vorticity is positive in both hemispheres. Candidate maxima are filtered using a closed contour vorticity test, and track retention is controlled by a minimum vorticity threshold.

`TempestExtremes` can identify several candidate tracks from the same forecast, so an additional selector retained one physically plausible track per deterministic product or ensemble member. Candidates were ranked using forecast-derived information, including duration, intensity measures, and track smoothness. The highest scoring track was retained. IBTrACS best track data were used as the reference in plotting and comparison, but not for track selection. Unless stated otherwise, NeuralGCM and IFS ensembles used 25 members, while ECMWF ensemble used 50 members.

3.3 Paper Reproduction

In this section, the case-specific methodological details are described.

3.3.1 Medium-Range Forecast Accuracy Evaluation

The medium-range forecast accuracy experiment tested whether the reported accuracy of NeuralGCM could be reproduced. Two NeuralGCM models were run; the 0.7° deterministic model and the 1.4° stochastic ensemble model. These were compared with physics-based forecasts, ECMWF HRES and ECMWF ensemble and data-driven forecasts, GraphCast and Pangu.

All forecasts were initialised twice per day, at 00 Coordinated Universal Time (UTC) and 12 UTC, for every day in 2020. This gave 732 forecast initialisations. Each forecast was run for 15 days with output saved every six hours, giving lead times from +6 hours to +360 hours with persistent forcings. The stochastic 1.4° NeuralGCM ensemble was made of 15 members, rather than the 50 members used in the original paper to limit computational cost. In contrast, the ECMWF ensemble forecast contained 50 ensemble members.

All reference forecasts were downloaded on a common 1.5° grid, while NeuralGCM forecasts were regridded to this grid before evaluation. Temporal metrics were calculated for 500 hPa geopotential height, 850 hPa temperature and 700 hPa specific humidity. Spatial metrics were calculated for 700 hPa specific humidity at forecast day 10, with below-surface regions masked using surface pressure. Beyond matching the original paper, the accuracy of these variables shows whether NeuralGCM reproduces broad atmospheric behaviour across circulation, temperature, and moisture fields.

All forecasts were evaluated using RMSE and RMSB. Ensemble forecasts were additionally evaluated using CRPS and spread-skill ratio. NeuralGCM, GraphCast and Pangu Weather were evaluated against ERA5, while ECMWF HRES and ECMWF ensemble were evaluated against the ECMWF operational analysis. Following the original paper convention, temporal RMSE, RMSB and CRPS were plotted relative to ECMWF ensemble, spatial RMSE and CRPS were plotted relative to a probabilistic climatology, and bias and spread-skill ratio were plotted directly.

3.3.2 Hurricane Laura Case Study

The Hurricane Laura case study used the same forecast products as Section 3.3.1, but took a subset of the forecast initialised at 12 UTC on 22nd August 2020, between Laura’s formation as a tropical depression on 20th August and landfall on 27th August. This directly reproduces the original NeuralGCM case study and tests whether the model captures moisture structure in a tropical cyclone, an atmospheric river, and the Intertropical Convergence Zone (ITCZ). Forecast fields were shown at lead times of one, five, and 10 days using 700 hPa specific humidity. ERA5 and ECMWF ensemble used the MSLP tracker, while 1.4° NeuralGCM ensemble used the vorticity tracker.

3.3.3 Annual Cyclone Tracks

This experiment tested NeuralGCM’s longer-term climate behaviour by assessing whether a continuous forecast produced a realistic annual distribution of tropical cyclone tracks. This used the deterministic 1.4° NeuralGCM model with dynamic forcings meaning SSTs and sea ice were updated every six hours. A single forecast was initialised at 00 UTC on 18th October 2019 and run continuously for 458 days until 00 UTC on 18th January 2021, with output saved every 12 hours. Tropical cyclone tracks were then calculated from 19th January 2020 to 17th January 2021.

For tracking, ERA5 was first regridded to the same 1.4° grid as NeuralGCM. As in the original paper, ERA5 tracks were identified with the MSLP tracker, while NeuralGCM tracks were identified with the vorticity tracker. No single cyclone selector was applied in this experiment, since the aim was to retain all tracks across the full analysis period rather than isolate one case.

After detection and stitching, the same post-tracking filtering was applied to both ERA5 and NeuralGCM tracks as stated in the original paper. Retained tracks were required to have genesis equatorward of 50° latitude, duration of at least 54 hours, at least 10 track points, at least 10 points over terrain below 150 m, and at least 10 points within 50° latitude.

3.4 Extension: Cyclone Case Studies

The NeuralGCM products analysed were deterministic NeuralGCM 0.7°, deterministic NeuralGCM 1.4°, and stochastic NeuralGCM 1.4°. ERA5 was used as the reference for dynamics and intensity diagnostics. For track comparison, ERA5, IBTrACS, IFS HRES, and IFS ensemble were also included. The NeuralGCM forecasts used dynamic forcings and outputs were saved every six hours. The methodology assessed three aspects of cyclone behaviour: track, intensity and the key dynamical challenge in each case: re-intensification, recurvature, and Fujiwhara interaction for Gulab and Shaheen, Faraji and Seroja-Odette respectively.

3.4.1 Cyclone Tracks

The track analysis addressed whether NeuralGCM captures cyclone tracks relative to the IBTrACS best track. It also tested the Faraji recurvature case. To keep the tracking method consistent across datasets, the vorticity tracker was used for both ERA5 and NeuralGCM products. The selected tracks were plotted against IBTrACS, ERA5, and IFS, then assessed visually, focusing on track curvature, termination, and ensemble spread.

For ensemble forecasts, an additional plot showed ensemble track density contours. These contours were derived from the spatial density of member tracks using a Gaussian kernel density estimate and plotted as probability mass contours enclosing 50%, 75%, and 90% of the estimated ensemble track density.

3.4.2 Cyclone Dynamics and Intensity

The dynamics and intensity analysis addressed whether NeuralGCM captures intensity measures, vertical coherence, and case-specific dynamical evolution relative to ERA5. Three diagnostics were produced. First, snapshot maps of 850 hPa cyclonic relative vorticity were used to diagnose horizontal vortex structure and tracker behaviour. By overlaying tracked positions, these maps show whether the tracker followed the intended circulation or shifted to a nearby vorticity maximum. Second, the 850 hPa geopotential height difference, defined as the minimum height difference between the vortex centre and the surrounding 500 to 1000 km environment, was used as an intensity measure. A larger negative difference indicates a deeper lower-tropospheric circulation. For Gulab and Shaheen, successful re-intensification would appear as two distinct negative minima, corresponding to the initial Gulab cyclone and later Shaheen re-intensification. Third, time-pressure vortex structure plots showed the maximum cyclonic relative vorticity within 500 km of the tracked centre from 850 to 100 hPa. Stronger and more vertically continuous vorticity indicates a more coherent vortex, while shallow or discontinuous vorticity indicates weaker vertical structure.

Additional diagnostics were applied to Seroja-Odette to assess the Fujiwhara interaction. These consisted of vortex separation through time, relative bearing through time, and midpoint-centred 850 hPa wind quiver plot. Together, these assess whether NeuralGCM captured the relative rotation and coupled flow structure of the two vortices.

4 Results

4.1 Paper reproduction

This section presents the results of the NeuralGCM paper reproduction. It first compares the reproduced medium-range forecast accuracy metrics with the original analysis in the paper, before presenting the reproduced Hurricane Laura case study and the annual tropical cyclone tracks.

4.1.1 Medium-Range Forecast Accuracy Evaluation

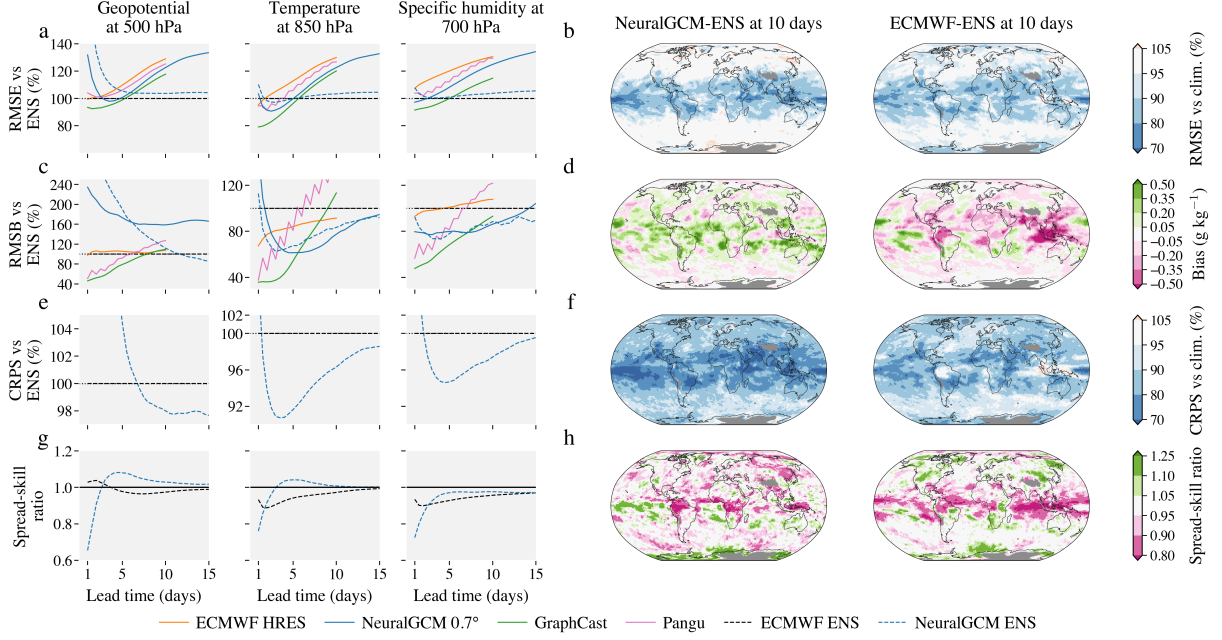


Figure 2: Forecast accuracy metrics for NeuralGCM and reference forecast systems. Panels (a), (c), (e) and (g) show metrics as a function of forecast lead time for 500 hPa geopotential, 850 hPa temperature and 700 hPa specific humidity. Panel (a) shows RMSE relative to ECMWF ENS, panel (c) shows RMSB relative to ECMWF ENS, and panel (e) shows CRPS relative to ECMWF ENS. Panel (g) shows the spread skill ratio for the ensemble forecasts, with the horizontal line marking a ratio of 1. Panels (b), (d), (f) and (h) show global maps for 700 hPa specific humidity at day 10. In each map row, NeuralGCM ensemble is shown on the left and ECMWF ensemble is shown on the right. Panel (b) shows RMSE relative to probabilistic climatology, panel (d) shows bias, panel (f) shows CRPS relative to probabilistic climatology, and panel (h) shows the spread skill ratio. Grey areas indicate regions where climatological surface pressure on average is below 700 hPa.

Figure 2 shows the reproduced medium-range forecast accuracy metrics for NeuralGCM and the reference forecasts. The left-hand panels show temporal RMSE, RMSB, CRPS and spread-skill ratio for 500 hPa geopotential height, 850 hPa temperature and 700 hPa specific humidity. For RMSE, RMSB and CRPS, values are plotted relative to ECMWF ENS, so values below 100% indicate lower error than the ECMWF ensemble reference. The spatial RMSE and CRPS maps are plotted relative to a probabilistic climatology, defined as a climatological ensemble with one member from each year between 1990 and 2019.

The reproduction captures the main qualitative result from the original paper, namely NeuralGCM provides competitive medium-range forecast performance, particularly for ensemble probabilistic metrics. The reproduced NeuralGCM ensemble remains close to, and mostly be-

low, ECMWF ENS in CRPS, although the y-axis spans only a narrow range around 100%, so the improvement is modest. This is notable because the reproduced NeuralGCM ensemble contains 15 members, compared with 50 in the original paper. The spread-skill ratio also remains close to one for much of the forecast window, indicating broadly consistent ensemble calibration.

However, the reproduced improvement is weaker than in the original paper. The clearest discrepancy occurs in the temporal evolution of the 500 hPa geopotential, where the reproduced NeuralGCM curves show a larger relative RMSE and RMSB and smaller CRPS improvement over ECMWF ensemble than in the original paper. The global forecast maps retain the broad patterns but with stronger contrasts in RMSE, bias, CRPS and spread-skill ratio. The main difference with the original paper is in RMSE against climatology (Figure 2b) where the reproduced maps show larger areas close to 100% of the climatology.

4.1.2 Hurricane Laura Case Study

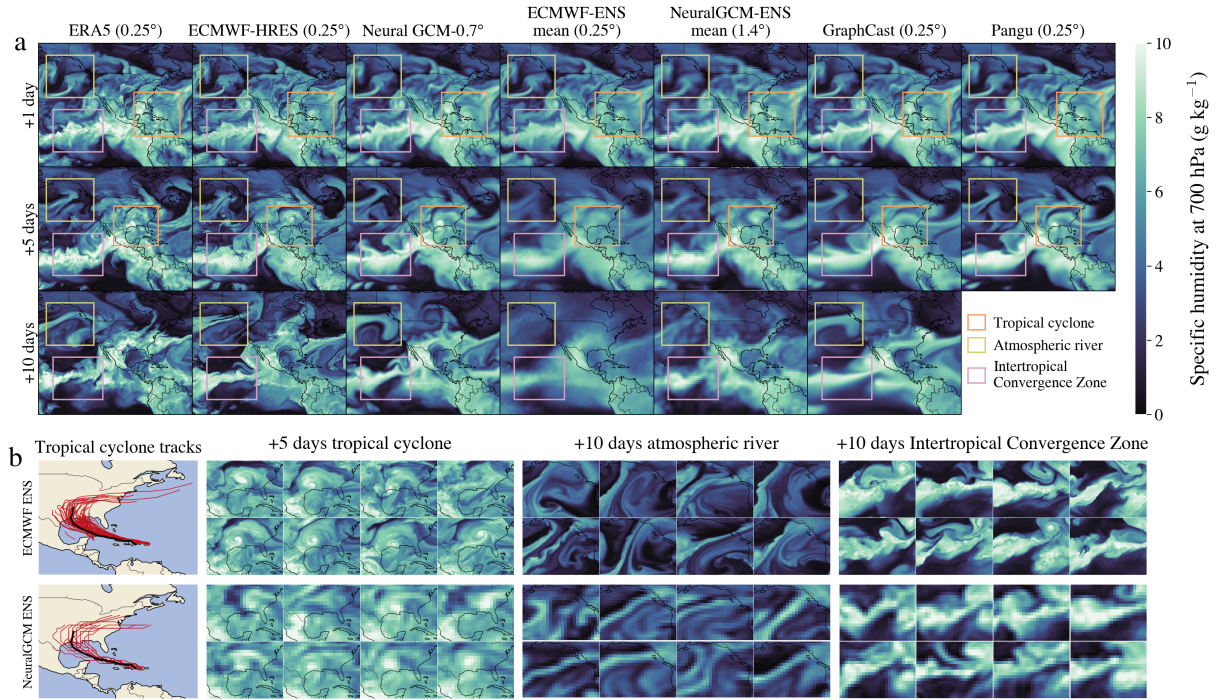


Figure 3: Medium-range Hurricane Laura case study. Panel (a) shows 700 hPa specific humidity for ERA5, ECMWF HRES, NeuralGCM 0.7°, ECMWF ensemble mean, NeuralGCM ensemble mean, GraphCast and Pangu at one, five and ten day lead times from the 22nd August 2020 12 UTC initialisation. Coloured boxes mark regions containing Hurricane Laura (orange), an atmospheric river (yellow) and the Intertropical Convergence Zone (ITCZ) (pink). Panel (b) compares ECMWF ensemble and NeuralGCM ensemble behaviour, showing Hurricane Laura tracks and selected ensemble member examples for the cyclone, atmospheric river and ITCZ regions. ECMWF ensemble contains 50 members, while NeuralGCM ensemble contains 15 members.

Figure 3a shows the humidity structures and tracks for Hurricane Laura. This figure shows that even at one day lead time, the ERA5 reference and physics-based ECMWF HRES retain sharper humidity features than the data-driven Pangu and GraphCast forecasts. This difference becomes more pronounced by five and ten days, when the data-driven products are visibly more diffuse in the Hurricane Laura moisture plume, the atmospheric river and the ITCZ. Compared to the data-driven models, the deterministic 0.7° NeuralGCM appears similarly smoothed to GraphCast and Pangu forecasts despite its coarser resolution. The ensemble mean forecasts are smoother still, especially for ECMWF ensemble, while the NeuralGCM ensemble mean is also

smoothed but still retains some humidity structure.

Figure 3b shows that NeuralGCM ensemble captures the broad path geometry of Hurricane Laura, although its member tracks are generally shorter than those in ECMWF ensemble. The selected NeuralGCM members reproduce regional humidity variability but are smoother than ECMWF ENS members, partly because NeuralGCM ENS is run on a coarser 1.4° grid. Despite using 15 members compared with 50 in ECMWF ENS, NeuralGCM ENS retains the large-scale moisture patterns associated with the cyclone, atmospheric river and ITCZ.

4.1.3 Annual Tropical Cyclone Tracks

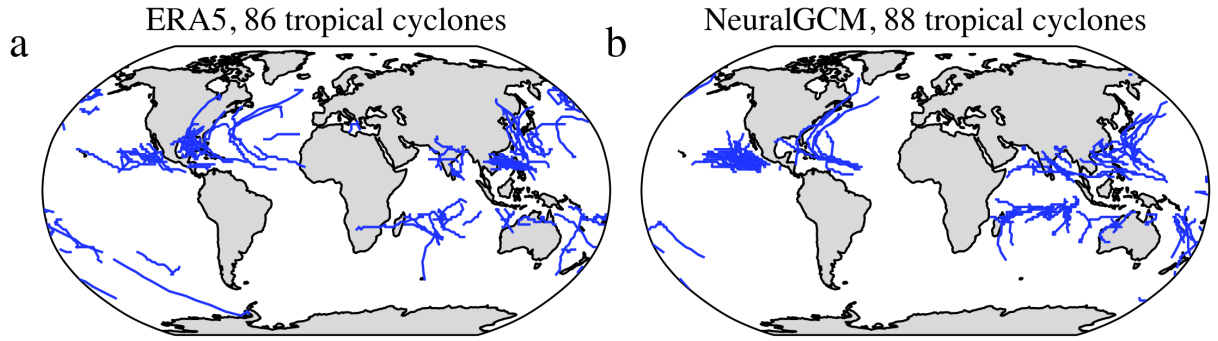


Figure 4: Tropical cyclone tracks from ERA5 and the deterministic 1.4° NeuralGCM forecast covering 19th January 2020 to 17th January 2021. Panel (a) shows filtered ERA5 tropical cyclone tracks, with 86 tracks detected. Panel (b) shows filtered NeuralGCM tropical cyclone tracks where 88 tracks were detected. ERA5 was regridded to a 1.4° Gaussian grid before computation and tracking.

Figure 4 shows tropical cyclone tracks from 19th January 2020 to 17th January 2021 for reference data, ERA5 and deterministic 1.4° NeuralGCM forecast. ERA5 produced 86 tropical cyclone tracks, while NeuralGCM produced 88. This is close to the original paper, where the corresponding counts were 86 and 83 respectively. The global maps capture the main regions of tropical cyclone activity, including the North Atlantic, western North Pacific, eastern North Pacific and parts of the Indian Ocean.

4.2 Extension

In this section the extension results are presented, first the tracks followed by the dynamics for each cyclone, Gulab and Shaheen, Faraji and Seroja in turn.

4.2.1 Cyclone Tracks

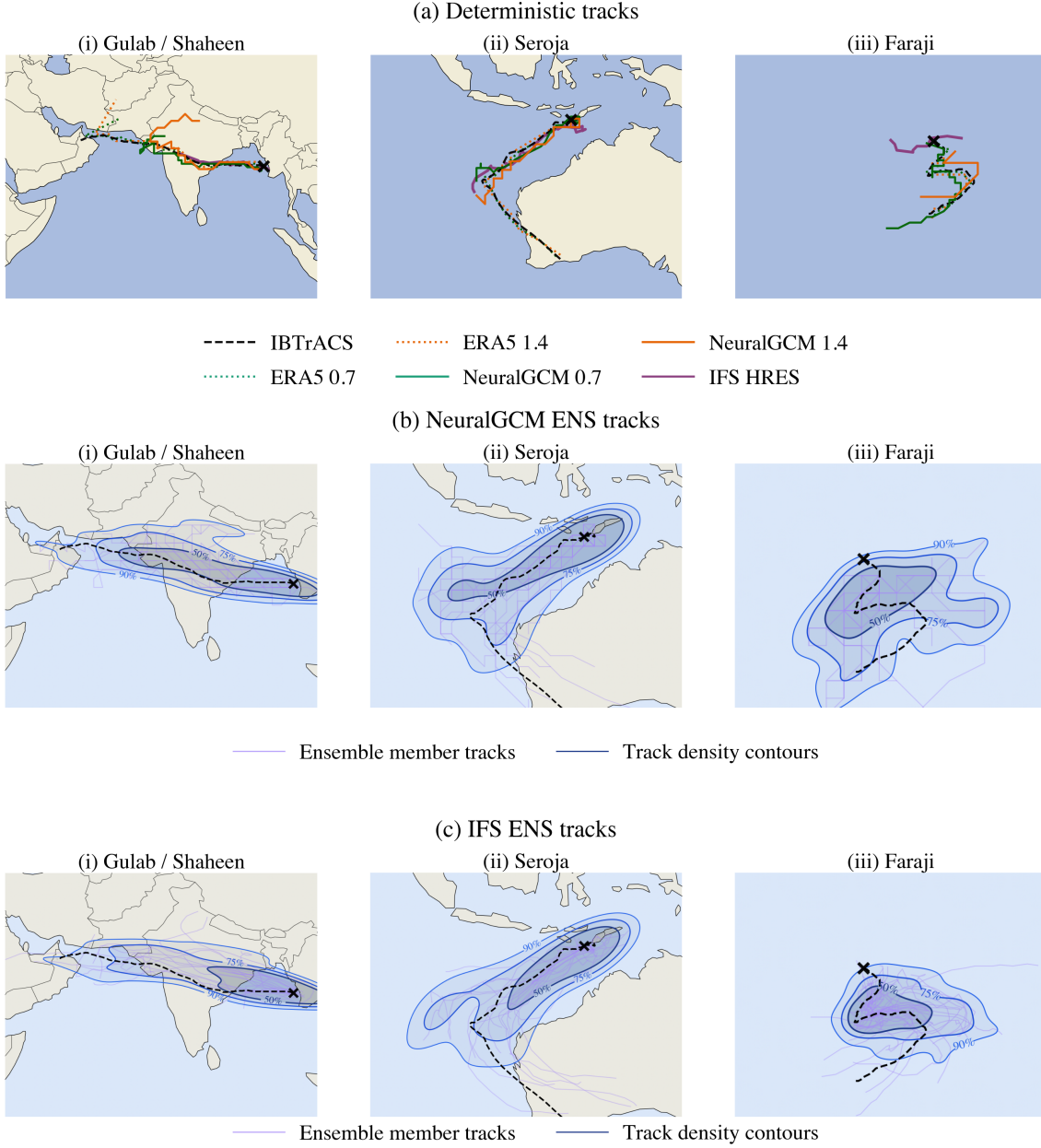


Figure 5: Cyclone track comparison for Gulab and Shaheen, Seroja, and Faraji. Panel (a) shows deterministic tracks from the IBTrACS best track (black dashed), ERA5 0.7° (green dotted), ERA5 1.4° (orange dotted), NeuralGCM 0.7° (green solid), NeuralGCM 1.4° (orange solid), and IFS HRES (purple solid). Panels (b) and (c) show ensemble member tracks in pale blue and track density contours in darker blue for the 25-member NeuralGCM ensemble and IFS ENS. The contours show enclosed probability mass from the spatial density of ensemble tracks. Black crosses mark the start point of the IBTrACS track.

Figure 5 shows the cyclone tracks for Gulab and Shaheen, Seroja and Faraji. ERA5 generally follows the IBTrACS best track closely, suggesting that the tracker identifies the main systems reliably. The late northward turn in the ERA5 Gulab and Shaheen is the main exception. This is likely a tracking artefact caused by switching to a nearby vorticity maximum.

NeuralGCM captures the broad path in all three cases, but with clear case-dependent errors. For Gulab and Shaheen, both deterministic runs follow Gulab but turn northward after re-emergence over the Arabian Sea, missing Shaheen's later westward motion. The NeuralGCM ensemble is closer to the full observed track, while the IFS ensemble is shorter and more tightly clustered. For Seroja, NeuralGCM and IFS show similar behaviour. All deterministic runs terminate near the Fujiwhara interaction, and both ensembles show a preference towards Odette at that point, more strongly in NeuralGCM. For Faraji, the deterministic NeuralGCM runs capture the recurvature, but the ensemble is broad and poorly defined. IFS shows the opposite pattern, with a poor deterministic track but an ensemble that is more centred on the initial recurvature, although it still misses the full observed evolution.

4.2.2 Cyclone Dynamics

Gulab and Shaheen

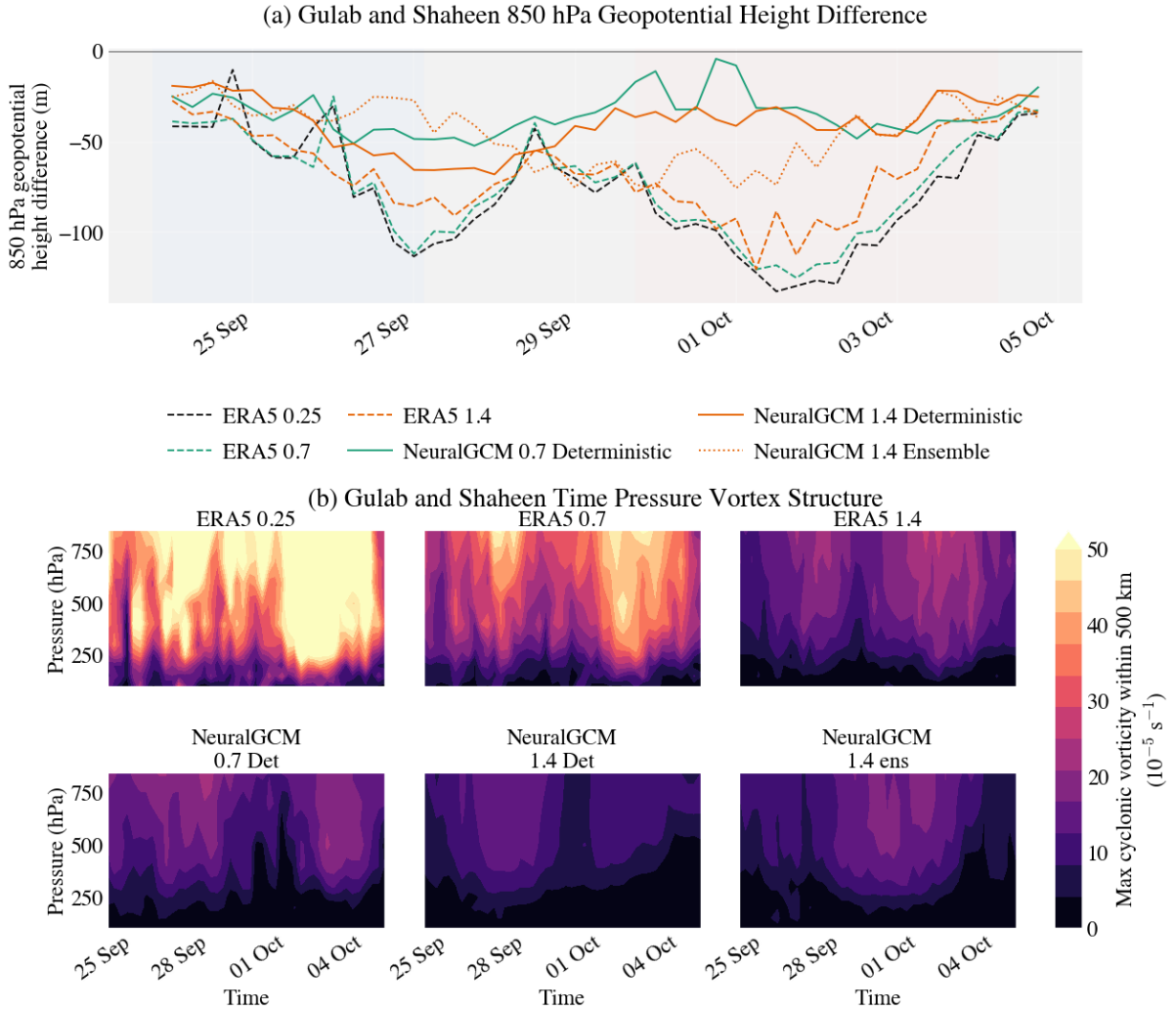


Figure 6: Evolution of the Gulab and Shaheen vortex structure in ERA5 and NeuralGCM. Panel (a) shows the 850 hPa geopotential height difference near the vortex centre from 24th September to 5th October. Panel (b) shows the time pressure vortex structure over the same period for ERA5 at 0.25°, 0.7°, and 1.4°, and for NeuralGCM 0.7° deterministic, 1.4° deterministic, and 1.4° ensemble runs.

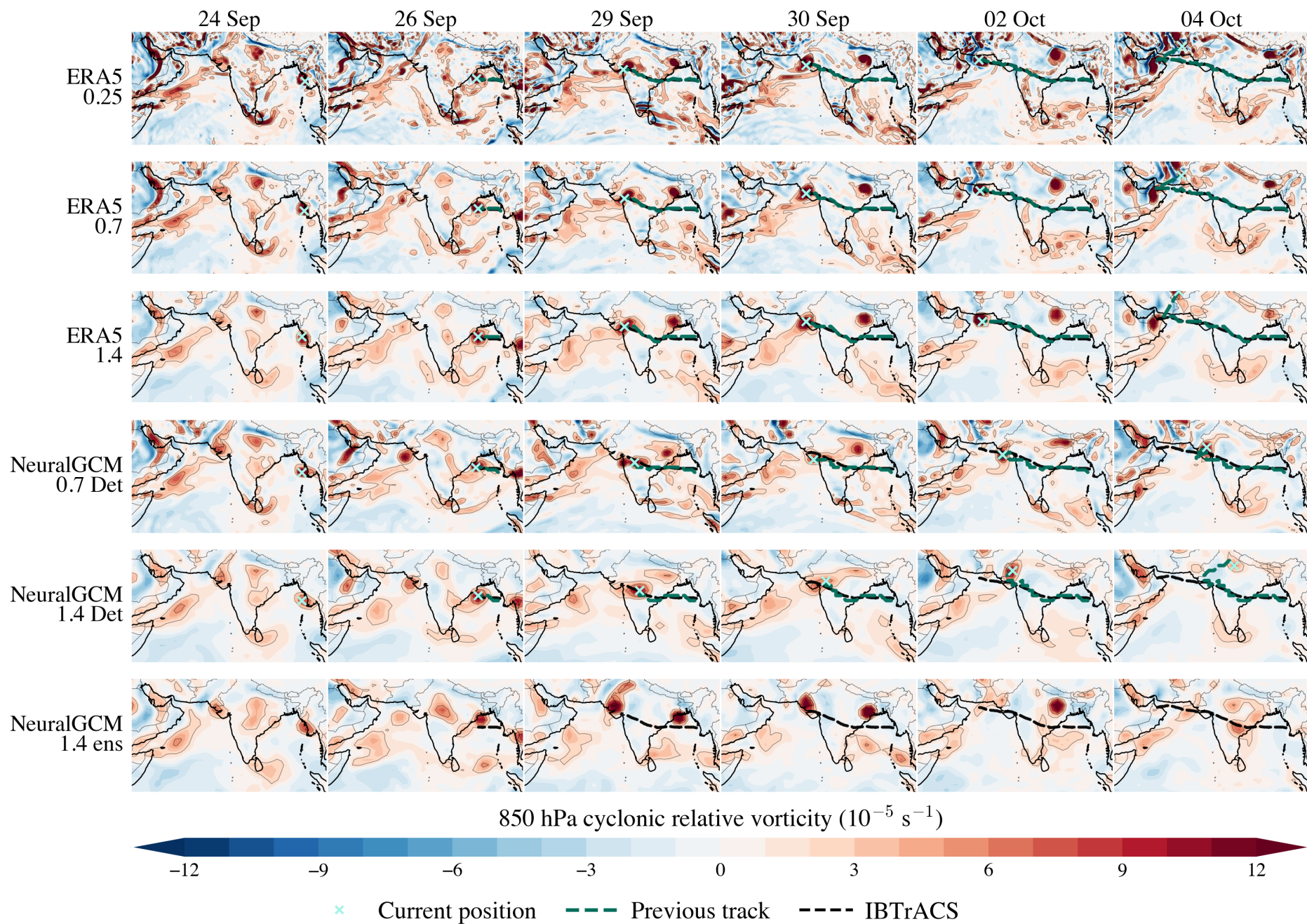


Figure 7: 850 hPa relative-vorticity snapshot panels for Gulab and Shaheen with tracked positions overlaid. The light green crosses show the current tracked positions, the green dashed lines show the previous model tracks, and the black dashed lines show the IBTrACS reference tracks. No previous track or current position is shown for the ensemble panels.

Figures 6 and 7 compare Gulab and Shaheen vortex evolution across ERA5 and NeuralGCM. The largest differences occur after Gulab makes landfall, weakens over land, and later re-intensifies over the Arabian Sea as Shaheen. In Figure 6a, ERA5 shows two clear 850 hPa geopotential height difference minima around 27th September and 2nd October, marking the initial Gulab cyclone and later Shaheen re-intensification. These minima are weaker in NeuralGCM 0.7°, weaker again in NeuralGCM 1.4°, and largely absent in the ensemble. Figure 6b shows the same pattern in vortex structure, with NeuralGCM producing a shallower and less vertically coherent vortex, especially after 29th September. Figure 7 supports the interpretation that the late northward turn in the ERA5 track is a tracking artefact, caused by the tracker shifting from the main vortex to a nearby vorticity maximum.

Faraji

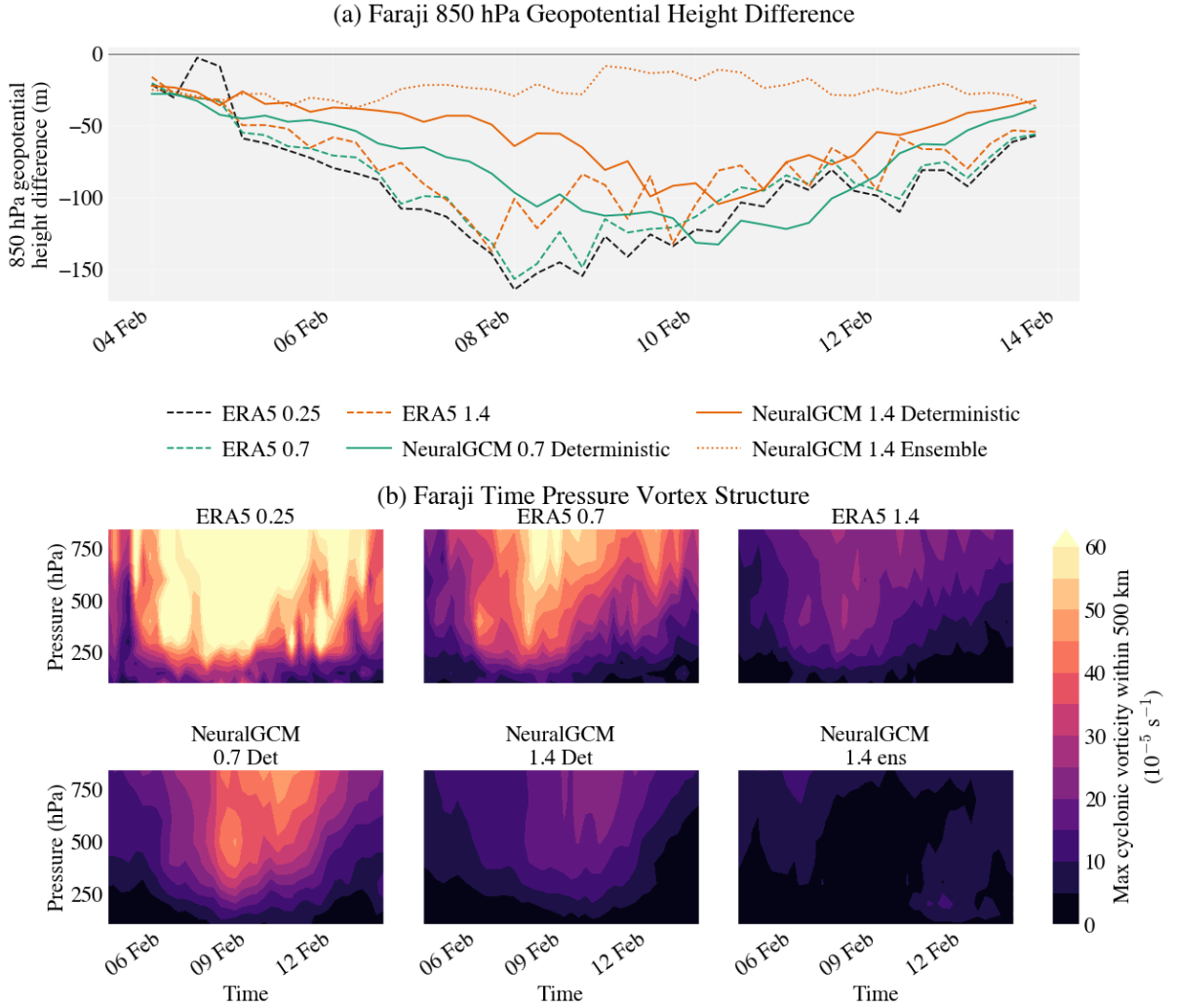


Figure 8: Evolution of Faraji vortex structure in ERA5 and NeuralGCM. Panel (a) shows the 850 hPa geopotential height difference near the vortex centre from 4th to 14th February. Panel (b) shows the time pressure vortex structure over the same period for ERA5 at 0.25°, 0.7°, and 1.4°, and for NeuralGCM 0.7° deterministic, 1.4° deterministic, and 1.4° ensemble runs. ERA5 shows a clear deepening and vertically coherent vortex through the intensification period, while NeuralGCM 0.7° deterministic captures this evolution most closely.

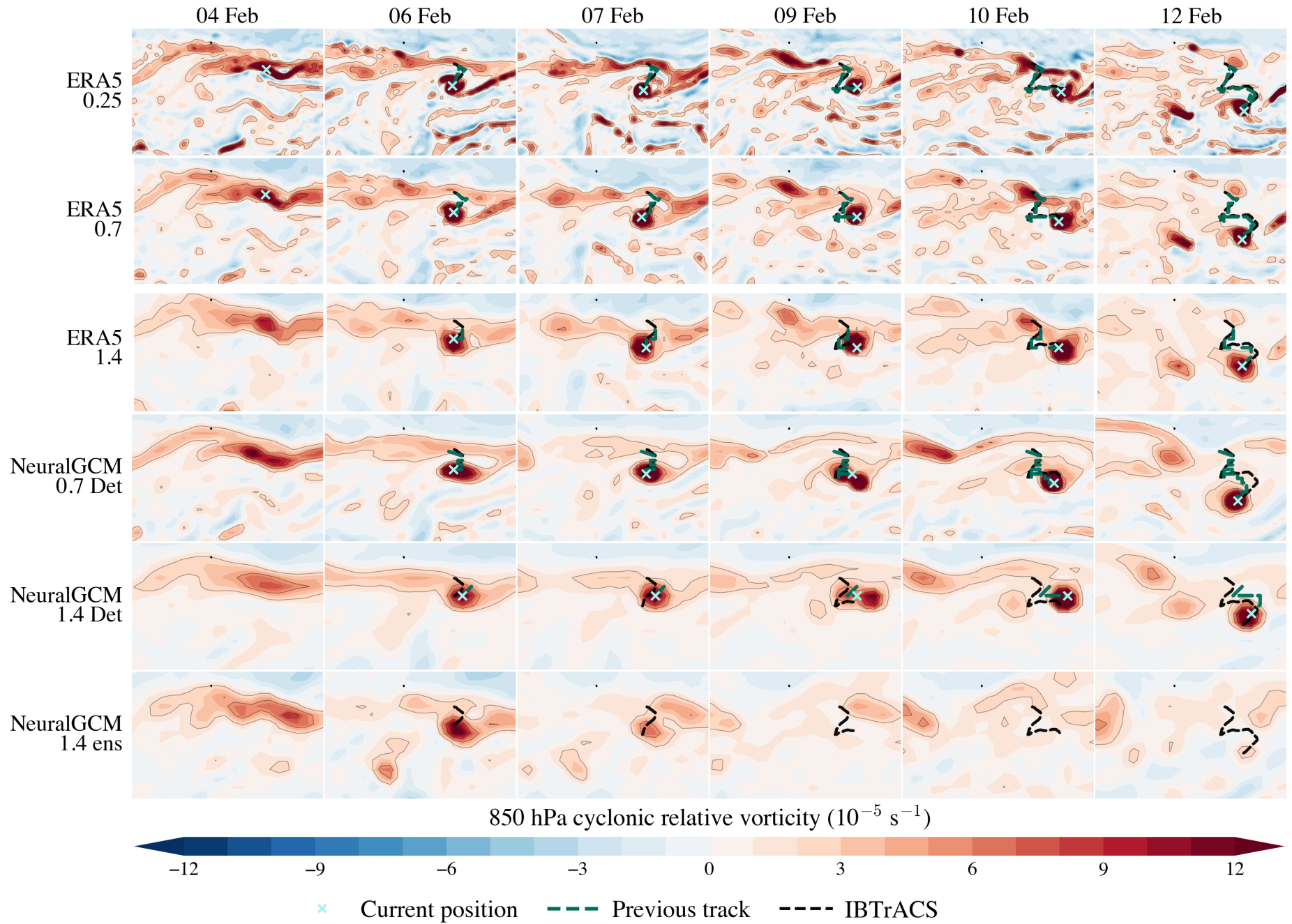


Figure 9: 850 hPa cyclonic relative vorticity snapshot panels for Faraji with tracked positions overlaid. The light green crosses show the current tracked positions, the green dashed lines show the previous model tracks, and the black dashed lines show the IBTrACS reference tracks. No previous track or current position is shown for the ensemble panels.

Figures 8 and 9 compare Faraji vortex evolution across ERA5 and NeuralGCM. Figure 8 shows that Faraji is reproduced most convincingly by deterministic NeuralGCM, especially at 0.7° . ERA5 shows strong deepening around 8th February, while NeuralGCM 0.7° captures a comparable but delayed minimum closer to 10th February. NeuralGCM 1.4° remains weaker throughout, and the ensemble largely misses the negative height difference. Figure 8b shows the same pattern in vortex structure, with NeuralGCM 0.7° retaining a coherent but weaker vortex, while the ensemble produces little vertically organised vorticity. Figure 9 supports this result, showing weaker cyclonic relative vorticity in the NeuralGCM ensemble than in ERA5.

Seroja

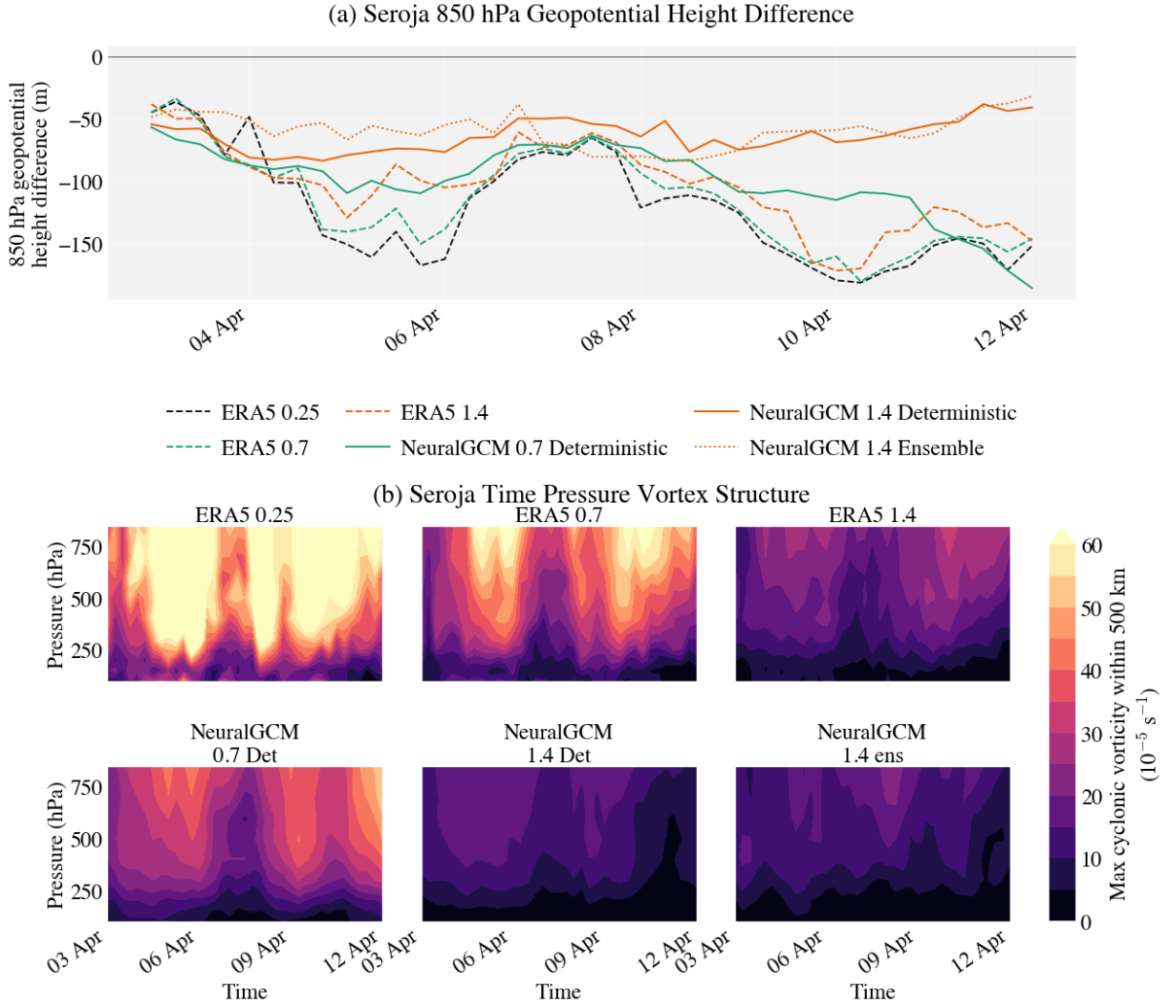


Figure 10: Evolution of Seroja vortex structure in ERA5 and NeuralGCM. Panel (a) shows the 850 hPa geopotential height difference near the vortex centre from 3rd to 12th April. Panel (b) shows the time pressure vortex structure over the same period for ERA5 at 0.25° , 0.7° , and 1.4° , and for NeuralGCM 0.7° deterministic, 1.4° deterministic, and 1.4° ensemble runs. ERA5 shows a deeper and more vertically coherent vortex during Seroja intensification, while the NeuralGCM products weaken markedly through the interaction period.

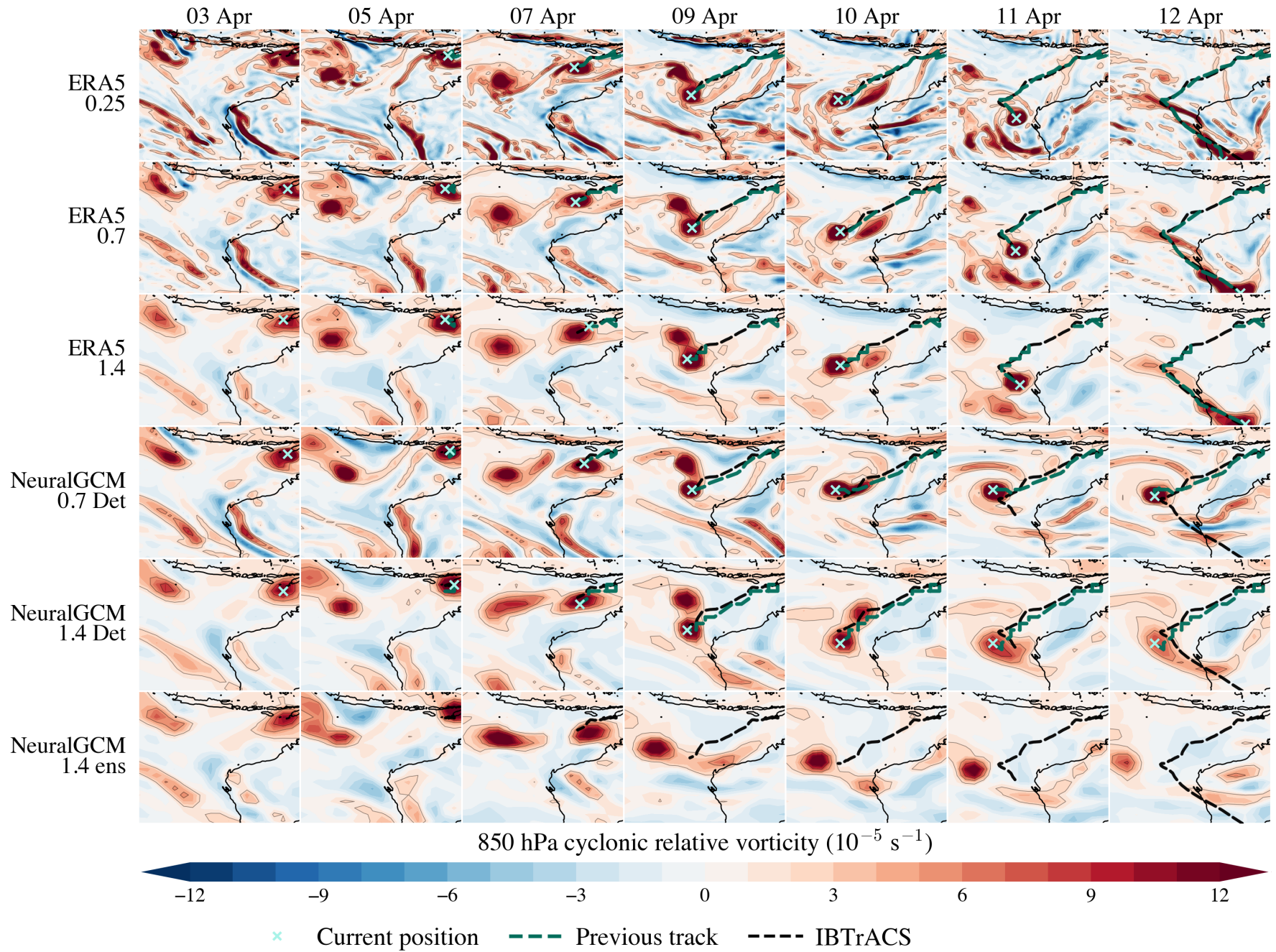


Figure 11: 850 hPa relative-vorticity snapshot panels for Seroja with tracked positions overlaid. The light green crosses show the current tracked positions, the green dashed lines show the previous model tracks, and the black dashed lines show the IBTrACS reference tracks. No previous track or current position is shown for the ensemble panels.

Figures 10 and 11 compare Seroja vortex evolution across ERA5 and NeuralGCM. Figure 10 shows that NeuralGCM struggles to capture the intensity of Seroja, especially during the Fujiwhara interaction with Odette. In panel (a), ERA5 shows two clear 850 hPa geopotential height difference minima around 5th and 10th April, corresponding to deepening before and after the interaction. NeuralGCM 0.7° partly captures this evolution, but with weaker and time-shifted deepening, while NeuralGCM 1.4° and the ensemble remain too shallow for most of the period. Panel (b) shows the same loss of vertical structure, with NeuralGCM producing weaker and shallower vortices than ERA5. Figure 11 shows that NeuralGCM forecasts lose cyclonic relative vorticity intensity after 10th April. By this stage, there is no clear vortex moving towards Australia for the tracker to follow, explaining why the tracks terminate at the corner.

The Fujiwhara effect, where two cyclones interact and rotate around a common centre, is further investigated in Figures 12 and 13.

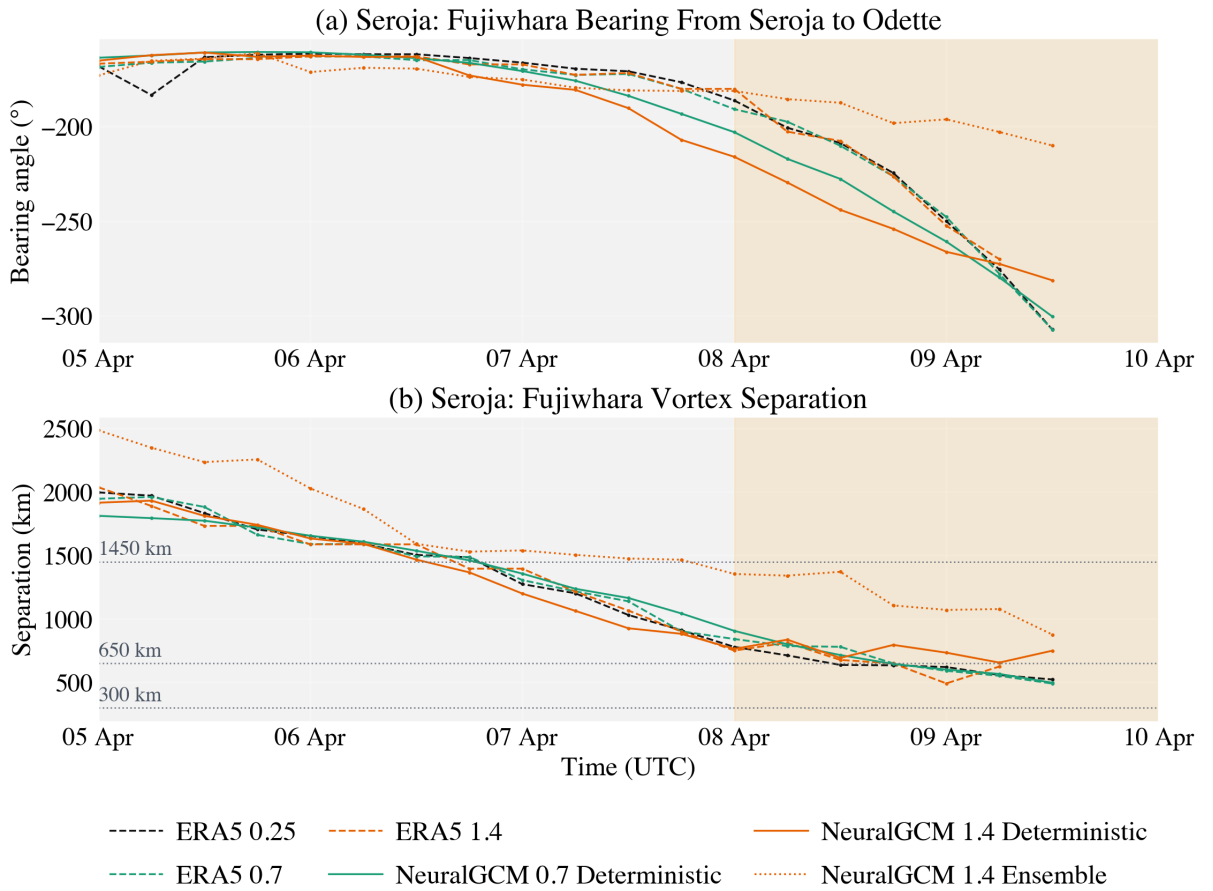


Figure 12: Fujiwhara interaction diagnostics for Seroja-Odette in ERA5 and NeuralGCM. Panel (a) shows the bearing from Seroja to Odette and panel (b) shows the vortex separation from 5th to 9th April. The dotted horizontal lines mark approximate separation thresholds for Fujiwhara interaction at 1450 km, stronger binary rotation at 650 km, and possible merger at 300 km.

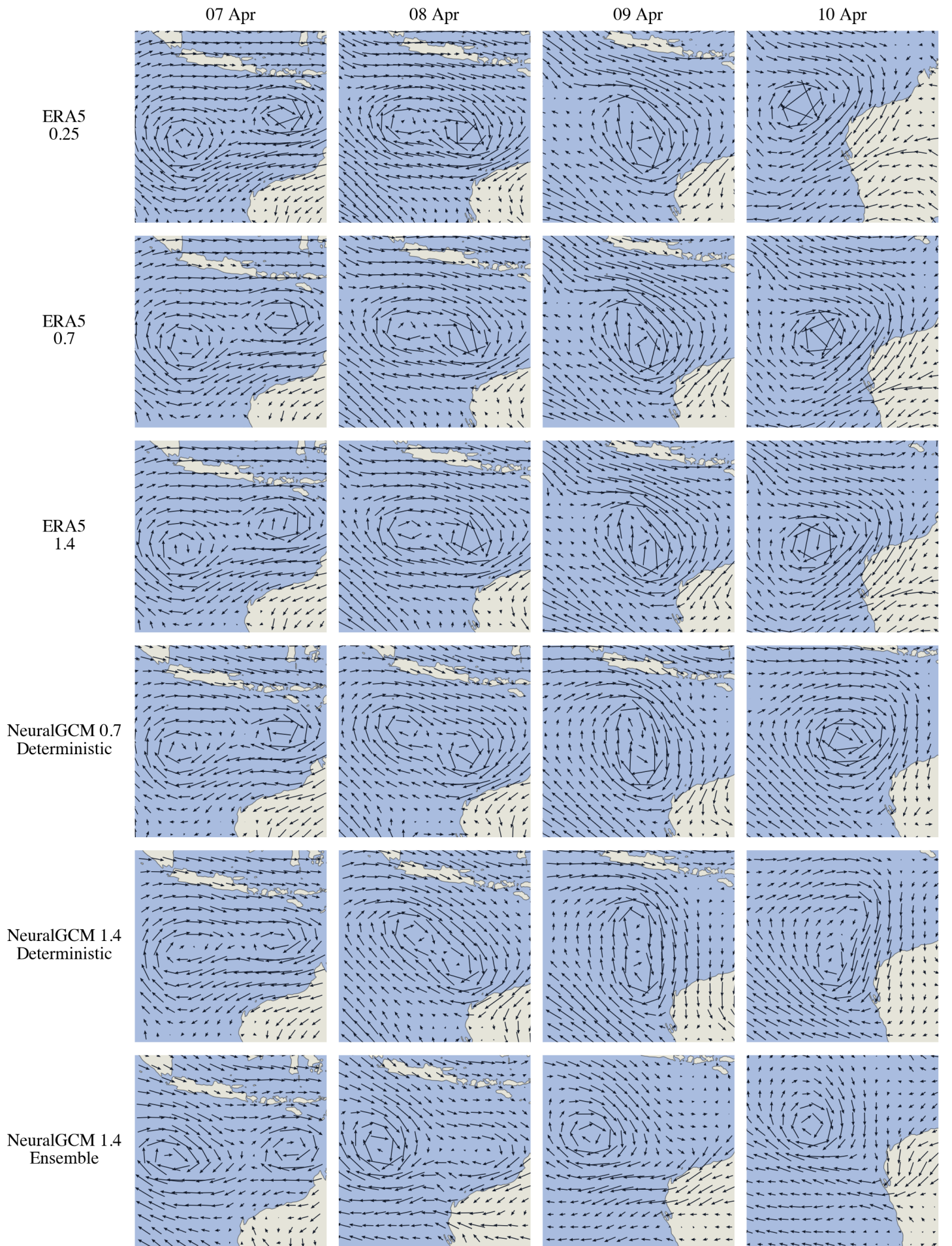


Figure 13: Midpoint-centred 850 hPa wind vectors for the Seroja-Odette Fujiwhara interaction on 7th to 10th April in ERA5 and NeuralGCM. Columns show the four dates and rows show ERA5 at 0.25°, 0.7°, and 1.4°, followed by NeuralGCM 0.7° deterministic, 1.4° deterministic, and 1.4° ensemble runs.

Figures 12 and 13 show the relative motion and 850 hPa wind structure of Seroja-Odette during the Fujiwhara interaction. In Figure 12a, ERA5 shows little bearing change until 7th April, followed by rapid clockwise rotation from about -180° to below -300° by 9th April. NeuralGCM 0.7° follows this evolution most closely, while NeuralGCM 1.4° deterministic rotates too quickly and the 1.4° ensemble changes bearing too slowly, remaining near -200° . In Figure 12b, ERA5 and the deterministic NeuralGCM runs reduce vortex separation from around 1800 to 2000 km on 5th April to below 650 km by around 8th April, then approach the 300 km close-interaction scale by 9th April. The 1.4° ensemble remains too widely separated for longer and enters the close-interaction regime too late.

Figure 13 supports this interpretation. ERA5 shows two distinct cyclonic circulations on 7th and 8th April, before the flow wraps into a shared rotating circulation by 9th and 10th April. NeuralGCM 0.7° reproduces this evolution most closely. NeuralGCM 1.4° deterministic captures the broad rotation but keeps the vortices more separated, while the 1.4° ensemble loses the second vortex after 8th April, showing no binary interaction.

5 Discussion

5.1 Paper reproduction

The reproduction recovered the original NeuralGCM conclusions, but not exactly. The medium-range evaluation reproduced the qualitative result that NeuralGCM is competitive on benchmark forecast accuracy, especially for NeuralGCM ensemble metrics. However, the reproduced metrics were weaker than in the original paper, with the clearest discrepancy in 500 hPa geopotential height. One possible source of this difference is pre-processing. The original paper used ERA5 pressure-level inputs, whereas this reproduction used ERA5 model-level data from CEDA and vertically interpolated them onto the required pressure levels. Since the paper does not fully specify the pre-processing and interpolation steps, small differences in vertical interpolation, surface pressure treatment or below-surface handling may have propagated into the initial model state. This is particularly relevant for geopotential height, which is sensitive to vertical coordinate treatment.

The global forecast maps reproduced the broad regional bias structure seen in the original paper, especially in the tropics, but with stronger contrasts. This suggests qualitative agreement but different spatial error magnitudes. Such sensitivity is plausible because tropical humidity is closely linked to convection, moisture organisation and cyclone development, which are affected by resolution, parametrised physics and pre-processing choices [44, 45].

The reproduced NeuralGCM ensemble remained broadly similar in shape to the original paper, but its probabilistic metrics were shifted upwards, indicating weaker ensemble performance. This may partly reflect the smaller 15-member ensemble used here, compared with 50 members in the original paper. Although the Hurricane Laura case study recovered the main qualitative behaviour, the smaller ensemble is likely to give a noisier estimate of forecast uncertainty and ensemble spread.

The annual tropical cyclone track experiment provides stronger evidence that the reproduction workflow was implemented consistently. ERA5 produced the same 86 filtered tracks as in the original paper, supporting the tracking and filtering method. NeuralGCM produced 88 tracks, compared with 83 in the original paper, but retained a similar broad global distribution. This suggests that the NeuralGCM count difference arises from the reproduced forecast rather than the post-processing workflow.

5.2 Extension

Overall, the extension results show that NeuralGCM reproduces cyclone tracks more successfully than cyclone structure and intensity, which are consistently weaker than in ERA5.

Although NeuralGCM is a hybrid physics-AI model, it appears to share a broader AI-model weakness, underestimating extreme-event intensity [35, 44, 45]. This may reflect three factors. (1) In NeuralGCM, sub-grid processes are handled by the ML component, while cyclone intensity depends strongly on convection and surface fluxes, which are difficult to resolve and learn directly. (2) The model is trained on ERA5 from 1979 to 2019, so extreme cyclone cases are limited. (3) Its loss functions optimise overall forecast accuracy rather than extremes specifically. Together, these factors may favour smoother predictions and weaker intensity representation. Despite the NeuralGCM ensemble’s stronger forecast metrics, it performs worse for intensity across all cases, likely because ensemble spread and smoothing can reduce CRPS by lowering average forecast errors while also smoothing the cyclone core and underestimating intensity. NeuralGCM fails to capture Gulab re-intensification into Shaheen. The vorticity snapshots

show this is not simply a tracking fault because after land crossing, NeuralGCM does not maintain a coherent westward-moving 850 hPa cyclonic vorticity maximum. Instead, the dominant remaining vorticity evolves northward, suggesting loss or displacement of the remnant vortex and possible errors in the surrounding flow. Studies note that Shaheen developed from Gulab's remnant circulation, supported by favourable Arabian Sea moisture, ocean heat content and wind-shear conditions [46, 47]. NeuralGCM therefore misses the remnant-vortex persistence and environmental evolution needed for re-intensification.

For Faraji, deterministic NeuralGCM captures the recurvature, but the stochastic ensemble tracks spread out and lose coherence. This suggests that small stochastic perturbations degrade the recurving pathway in this case. IFS shows the opposite pattern: the deterministic forecast misses the recurvature, while the ensemble partly captures it, although imperfectly.

The Fujiwhara effect is best captured by NeuralGCM 0.7° , although it remains weaker than ERA5 0.7° . The interaction requires two coherent vortices with the correct separation, relative strength and mutual advection. NeuralGCM 1.4° deterministic captures only a weakened interaction, while the 1.4° ensemble shows little to no coherent Fujiwhara interaction. This suggests that coarser resolution and stochastic member spread smooth or displace the vortices enough for the binary interaction to break down.

These extension results have two limitations. First, they use three selected cases rather than a large sample, so they identify failure modes rather than general performance. Second, the diagnostics show where NeuralGCM fails but do not isolate the physical cause, which future work should address.

6 Conclusion

This project evaluated NeuralGCM as a hybrid physics-AI weather and climate model by testing both benchmark forecast performance and physically meaningful tropical cyclone behaviour. The reproduction broadly supports the original NeuralGCM paper: NeuralGCM produced competitive medium-range forecast performance, especially for ensemble probabilistic metrics, qualitatively reproduced the Hurricane Laura case study, and generated a realistic annual distribution of tropical cyclone tracks. However, exact reproduction was not achieved. The clearest discrepancy was in the 500 hPa geopotential height metrics, suggesting sensitivity to implementation details such as pre-processing, interpolation and ensemble size.

The extension case studies gave mixed answers to the three aims. NeuralGCM often captured the broad cyclone path relative to the IBTrACS best track, with the clearest success in deterministic Faraji forecasts, but Gulab and Shaheen missed the later westward Shaheen motion and Seroja tracks degraded during or shortly after the Fujiwhara interaction. NeuralGCM also did not reliably reproduce the intensity measures or vertical coherence seen in ERA5, producing weaker 850 hPa geopotential height differences and shallower cyclonic relative vorticity structures than the ERA5 reference. For the key dynamical challenges, it missed Shaheen re-intensification, represented Faraji recurvature well in the deterministic forecasts but not in the stochastic ensemble, and captured the Seroja-Odette Fujiwhara interaction most convincingly at 0.7° , while the interaction was weak or absent in the 1.4° configurations.

Future work should test whether these weaknesses hold across a larger sample of storms, basins, lead times and model resolutions. More targeted diagnostics could investigate steering flow, vortex persistence, and the processes that strengthen or weaken 850 hPa cyclonic relative vorticity. NeuralGCM should also be assessed further as a climate model through longer forecasts that test model drift and stability.

The impact of this work is twofold. First, it provides a reproducibility test for NeuralGCM as an early hybrid physics-AI model, distinguishing broad agreement with the original paper from exact numerical replication. Second, it extends NeuralGCM evaluation beyond reproduction to more original tropical cyclone case studies, especially the Seroja-Odette Fujiwhara interaction in the Southern Hemisphere. To the best of the author’s knowledge, this is the first investigation of this behaviour in NeuralGCM or comparable data-driven weather models. Overall, the project shows both the promise and the current limits of NeuralGCM: it is a strong hybrid physics-AI forecasting framework, but its tropical cyclone performance remains less reliable when forecasts depend on cyclone-scale structure, intensity and complex vortex interaction.

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A AI Declaration

Once a full draft was finished, AI was used for proofreading, minor wording improvements. Codex was used to assist with coding, as specified in the repository README.